

Quiz 2

NAME: _____

Q.1. [5 points] Simplify the following expression (a and b can be any real numbers):

$$\frac{\sqrt{a^4b^6}}{\sqrt{a^{12}b^8}}$$

We can combine the radicals into a single radical and then combine/ cancel like powers via the quotient rule for exponents:

$$\begin{aligned} & \sqrt{\frac{a^4b^6}{a^{12}b^8}} \\ &= \sqrt{a^{4-12}b^{6-8}} \\ &= \sqrt{a^{-8}b^{-2}} \end{aligned}$$

Now we remember that even roots can cancel even powers, but we need to introduce absolute value symbols:

$$= |a^{-4}b^{-1}|$$

Since a^{-4} is an even power of a , whatever number it may be it is guaranteed to be nonnegative, so you technically don't need an absolute value on that term. So the following answer is also correct:

$$= a^{-4}|b^{-1}|$$

You may also have written your answers as reciprocals of positive powers:

$$= \left| \frac{1}{a^4b} \right| \text{ OR } \frac{1}{a^4|b|}$$

Q.2. [5 points] Fully factor the following polynomial:

$$x^2 - y^2 - 6x - 9$$

TYPO IN QUESTION (should have been $x^2 - y^2 - 6y - 9$) QUESTION CANCELLED

Solution to intended question:

$$\begin{aligned} & x^2 - y^2 - 6y - 9 \\ &= x^2 - (y^2 + 6y + 9) \\ &= x^2 - (y + 3)^2 \\ &= (x - (y + 3)) \cdot (x + (y + 3)) \end{aligned}$$

$$= (x - y - 3) \cdot (x + y + 3)$$

Q.3. [5 points] Simplify the following rational expression:

$$\frac{x^3 - 2x^2}{x^2 - 3x - 10}$$

Let's start by factoring the numerator and denominator. $x^3 - 2x^2$ has GCF = x^2 , so $x^3 - 2x^2$ factors as $x^2(x - 2)$ and this happens to be a full factorization.

The denominator is a trinomial in one variable, so it makes sense to try our p and q method. Here $a = 1, b = -3, c = -10$, so $ac = -10$ and we may take $p = -5, q = 2$:

$$\begin{aligned} x^2 - 3x - 10 &= x^2 - 5x + 2x - 10 \\ &= x(x - 5) + 2(x - 5) = (x + 2)(x - 5) \end{aligned}$$

So our rational expression fully factored is:

$$\frac{x(x - 2)}{(x + 2)(x - 5)}$$

Do we see any common reduced factors we can cancel? No! So this is our final answer (with working present), and no reduction is possible.

$$\boxed{\frac{x^3 - 2x^2}{x^2 - 3x - 10}} \quad \text{OR} \quad \boxed{\frac{x(x - 2)}{(x + 2)(x - 5)}}$$

Q.4. [5 points] Simplify the following combination of rational expressions:

$$\frac{1}{x + 1} - \frac{1}{x + 2} + \frac{3}{(x + 1)(x + 2)}$$

To add/ subtract fractions, we first need to give them all the same denominator. The LCD in this problem is $(x + 1)(x + 2)$.

Giving each fraction that same denominator:

$$\begin{aligned} \frac{1}{x + 1} &= \frac{1}{x + 1} \cdot \frac{x + 2}{x + 2} = \frac{x + 2}{(x + 1)(x + 2)} \\ \frac{1}{x + 2} &= \frac{1}{x + 2} \cdot \frac{x + 1}{x + 1} = \frac{x + 1}{(x + 1)(x + 2)} \end{aligned}$$

And the last fraction already has the desired denominator, so we leave it as is, giving us:

$$\begin{aligned} &= \frac{x + 2}{(x + 1)(x + 2)} - \frac{x + 1}{(x + 1)(x + 2)} + \frac{3}{(x + 1)(x + 2)} \\ &= \frac{x + 2 - (x + 1) + 3}{(x + 1)(x + 2)} \\ &= \frac{x + 2 - x - 1 + 3}{(x + 1)(x + 2)} \end{aligned}$$

$$= \frac{4}{(x+1)(x+2)}$$

BONUS: Rationalize the denominator:

$$\frac{1}{\sqrt[3]{a} - \sqrt[3]{b}}$$

We see a difference of cube roots in the denominator, so should be thinking about using the difference of cubes formula to help us rationalize this.

Plugging $\sqrt[3]{a}$ and $\sqrt[3]{b}$ into the difference of cubes formula gives us:

$$(\sqrt[3]{a})^3 - (\sqrt[3]{b})^3 = (\sqrt[3]{a} - \sqrt[3]{b})(\sqrt[3]{a}^2 + \sqrt[3]{a}\sqrt[3]{b} + (\sqrt[3]{b})^2)$$

$$\Rightarrow a - b = (\sqrt[3]{a} - \sqrt[3]{b})(\sqrt[3]{a^2} + \sqrt[3]{a}\sqrt[3]{b} + \sqrt[3]{b^2})$$

So instead of a simple "conjugate", we should multiply and divide the fraction by $\sqrt[3]{a^2} + \sqrt[3]{a}\sqrt[3]{b} + \sqrt[3]{b^2}$:

$$\begin{aligned} \frac{1}{\sqrt[3]{a} - \sqrt[3]{b}} &= \frac{1}{\sqrt[3]{a} - \sqrt[3]{b}} \cdot \frac{\sqrt[3]{a^2} + \sqrt[3]{a}\sqrt[3]{b} + \sqrt[3]{b^2}}{\sqrt[3]{a^2} + \sqrt[3]{a}\sqrt[3]{b} + \sqrt[3]{b^2}} \\ &= \frac{\sqrt[3]{a^2} + \sqrt[3]{a}\sqrt[3]{b} + \sqrt[3]{b^2}}{a - b} \end{aligned}$$